Non-Invasive Prenatal Screening

Genolyx

Patient Name:	Jane Doe	**Sample Number:**	S-123456
Date of Birth:	1990-01-01	**Physician:**	Dr. Smith
Gestational Age:	12 Weeks 3 Days	**Hospital/Clinic:**	General Hospital
Pregnancy Type:	Singleton	**Collection Date:**	2023-10-01
Medical Record Number (MRN):	MRN-123456	**Report Date:**	2023-10-05
Indication:	Advanced Maternal Age	**Order ID:**	ORD-7890

Overall Result: Low Risk

Predicted Gender: **Female**

Fetal Fraction: 10.5%

CHROMOSOMAL ANEUPLOIDIES

Condition	Result	Condition	Result
Trisomy 21 (Down Syndrome)	**Low Risk**	Trisomy 9	**Low Risk**
Trisomy 18 (Edwards Syndrome)	**Low Risk**	Trisomy 16	**Low Risk**
Trisomy 13 (Patau Syndrome)	**Low Risk**	Trisomy 22	**Low Risk**
All Other Autosomal Trisomy	**Low Risk**		
XO (Turner Syndrome)	**Low Risk**	XXX (Trisomy X)	**Low Risk**
XXY (Klinefelter Syndrome)	**Low Risk**	XYY (Jacob Syndrome)	**Low Risk**

MICRODELETIONS & DUPLICATIONS

Condition	Result	Condition	Result
1p36 deletion syndrome	**Low Risk**	Wolf-Hirschhorn syndrome	**Low Risk**
Williams-Beuren syndrome	**Low Risk**	Prader-willi/Angelman syndrome	**Low Risk**
Cri Du Chat syndrome	**Low Risk**	Jacobsen syndrome	**Low Risk**
DiGeorge syndrome (22q11.2)	**Low Risk**	2q33.1 deletion syndrome	**Low Risk**
Other microdeletions/duplications	**Low Risk**		

INTERPRETATION

No aneuploidies or microdeletions detected. Fetal fraction is adequate. No aneuploidies or microdeletions detected. Fetal fraction is adequate. No aneuploidies or microdeletions detected. Fetal fraction is adequate. No aneuploidies or microdeletions detected. Fetal fraction is adequate. No aneuploidies or microdeletions detected. Fetal fraction is adequate.

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PERFORMANCE METRICS

Condition	Sensitivity	Specificity	PPV
Trisomy 21	99.72%	>99%	98.72%
Trisomy 18	99.04%	>99%	96.49%
Trisomy 13	98.84%	>99%	85.71%
Monosomy X	>99%	>99%	76%
XXX	>99%	>99%	>99%
XXY	>99%	>99%	>99%
XYY	>99%	>99%	>99%
T9, T16, T22	>99%	>99%	50-99%
All Other Trisomy	>99%	>99%	NA
Microdeletions/duplications	20-99%	>99%	NA

Limitations: This test is designed to screen for fetal chromosome abnormalities in singleton pregnancies that are at least 10 weeks gestation, as estimated by last menstrual period. Microdeletion sensitivity is dependent on size and fetal fraction; the minimum detectable size is 0.5 MB at 4% fetal fraction. However, it is important to note that this test does not rule out the possibility of other chromosomal or subchromosomal abnormalities, birth defects, or other conditions. Additionally, this test is not intended to identify pregnancies at risk for open neural tube defects. A negative test result does not mean that there are no chromosomal abnormalities present. There is a small chance that the test results may not reflect the chromosomes of the fetus, but may instead reflect chromosomal changes of the placenta or of the mother.

Methods: UCL laboratory performed low pass whole genome sequencing from extracted cfDNA. The sequencing data was validated in-house at UCL laboratory. PPV performance was calculated based on amniocentesis and QF-PCR results. Due to the insufficient number of amniocentesis samples, Sensitivity was calculated by estimating true positives based on PPV.

References:

1. Anthony R et al., Noninvasive prenatal screening for fetal aneuploidy 2016 update: a position statement of the American College of Medical Genetics and Genomics. Genet Med 2016; 13:1056-65.
2. Bianchi DW et al., DNA sequencing versus standard prenatal aneuploidy screening. N Engl J Med. 2014; 370:799-808.
3. YM Dennis Lo, RWK Chiu, Noninvasive prenatal diagnosis of fetal chromosomal aneuploidies by maternal plasma nucleic acid analysis; Clinical chemistry, 2008.
4. Chen, M., et al., Validation of fetal DNA fraction estimation and its application in noninvasive prenatal testing for aneuploidy detection in multiple pregnancies. Prenatal Diagnosis, 2019. 39(13): p. 1273-1282.

MICRODELETIONS/DUPLICATIONS SYNDROMES

Condition	Condition	Condition	Condition
1p36 terminal region (includes GABRD) gain	Partial trisomy 5p	Partial trisomy 12p	17q11.2 recurrent region (includes NF1) gain
1q21.1 recurrent region (distal, BP3-BP4) (includes GJA5) loss	Partial monosomy 5q	Partial trisomy 12q	17q11.2 recurrent region (includes NF1) loss
1q21.1 recurrent region (distal, BP3-BP4) (includes GJA5) gain	Partial trisomy 5q	Monosomy 13q14	17q12 recurrent (RCAD syndrome) region (includes HNF1B) gain
1q43-q44 terminal region (includes AKT3)	6q24 region (includes PLAGL1)	Monosomy 13q21-qter	17q12 recurrent (RCAD syndrome) region (includes HNF1B) loss
Partial monosomy 1p	Partial trisomy 6p	Trisomy 13cen-q14	17q21.31 recurrent region (includes KANSL1)
Monosomy 1q21-q32	Partial monosomy 6q	Trisomy 13q21-qter	17q23.1-q23.2 recurrent region (includes TBX2, TBX4) gain
Monosomy 1q42-qter	Partial trisomy 6q	14q11.2 region including CHD8 and SUPT16H	17q23.1-q23.2 recurrent region (includes TBX2, TBX4) loss
Trisomy 1q23-qter	7p22.1 region (includes ACTB)	DLK1-MEG3 Intergenic Region	SOX9 upstream enhancer region
2p15-p16.1 region (includes BCL11A)	7q11.23 recurrent (Williams-Beuren syndrome) region (includes ELN) gain	Partial trisomy 14	Trisomy 17pter-q21
2p24.3 MYCN-DDX1 duplication region	7q11.23 recurrent distal region (includes HIP1, YWHAG)	Trisomy 14q24-qter	Partial monosomy 18p
2q11.2 recurrent region (includes ARID5A, TMEM127)	7q36.3 ZRS (SHH cis-regulatory) duplication region (within LMBR1 intron 5)	15q11.2-q13 recurrent (PWS/AS) region (Class I, BP1-BP3 and Class II, BP2-BP3) gain	Partial tetrasomy 18p
2q13 recurrent region (distal) (includes BCL2L11) gain	Partial monosomy 7p	15q13.3 recurrent region (BP4-BP5) (includes CHRNA7)	Trisomy 18pter-q12
2q13 recurrent region (distal) (includes BCL2L11) loss	Partial trisomy 7p	15q13.3 recurrent region (D-CHRNA7 to BP5) (includes CHRNA7, OTUD7A)	Monosomy 18q21-qter
Partial monosomy 2p	Monosomy 7q32-qter	15q24 recurrent region (LCR A-LCR C)	Trisomy 18q12-qter
Partial trisomy 2p	Trisomy 7q21-q31	15q24 recurrent region (LCR A-LCR D) (includes SIN3A)	Partial monosomy 20p
Partial monosomy 2q	Trisomy 7q32-qter	15q24 recurrent region (LCR C-LCR D) (includes SIN3A)	Trisomy 20p
Partial trisomy 2q	8p23.1 recurrent region (includes GATA4) gain	15q25.2 recurrent region (proximal, LCR-A/B-C) (includes RPS17)	Partial monosomy 21
3q24 Region (includes ZIC1)	8p23.1 recurrent region (includes GATA4) loss	Supernumerary inv dup (15)	22q11.2 recurrent (DGS) region (proximal, A-B, A-C, or A-D) (includes TBX1) gain
3q29 recurrent region (includes DLG1)	Partial trisomy 8q	Trisomy 15q22-qter	22q11.2 recurrent region (central, B-D or C-D) (includes CRKL)
Monosomy 3p11-p21	Partial monosomy 9p	16p11.2 recurrent region (distal, BP2-BP3) (includes SH2B1)	22q11.2 recurrent region (distal type I, D-E or D-F)
Monosomy 3p25-pter	Partial trisomy 9p	16p11.2 recurrent region (proximal, BP4-BP5) (includes TBX6) gain	22q11.2 recurrent region (distal type III, D-G, D-H) (includes SMARCB1)
Partial trisomy 3p	Partial monosomy 9q	16p11.2 recurrent region (proximal, BP4-BP5) (includes TBX6) loss	22q11.2 recurrent region (distal type III, E-H or F-H) (includes SMARCB1)
Partial monosomy 3q	Partial trisomy 9q	16p12.2 recurrent region (proximal) (includes EEF2K, CDR2)	22q11.2 recurrent region (distal type III, F-G) (includes SMARCB1)
Partial trisomy 3q	10q22.3-q23.2 recurrent region (includes BMPR1A)	16p13.11 recurrent region (BP1-BP3, BP2-BP3, or BP2-BP4) (includes MYH11) gain	22q11.21 recurrent (CES) region (includes CECR2)
Partial monosomy 3p	Partial monosomy 10p	16p13.11 recurrent region (BP1-BP3, BP2-BP3, or BP2-BP4) (includes MYH11) loss	Xp22.31 recurrent region (includes STS)
4p16.3 terminal (Wolf-Hirschhorn syndrome) region gain	Partial trisomy 10p	16p13.3 region (includes CREBBP)	Xp21.2 region (includes NROB1)
Partial monosomy 4p	Partial monosomy 10q	Partial trisomy 16p	Xp11.23 region (includes MAOA and MAOB)
Partial trisomy 4p	Partial trisomy 10q	Partial monosomy 16q	Xp11.22-p11.23 recurrent region (includes SHROOM4)
Monosomy 4q21-q31	11p11.2 (Potocki-Shaffer syndrome) region (includes ALX4, EXT2)	Partial trisomy 16q	Xp11.22 region (includes HUWE1)
Monosomy 4q31-qter	11p13 (WAGR syndrome) region	17p11.2 recurrent (SMS/PLS) region (includes RAI1) gain	Xq25 region (includes STAG2) gain
Partial trisomy 4q	11q13.2-q13.4 recurrent region (includes SHANK2, FGF3)	17p11.2 recurrent (SMS/PLS) region (includes RAI1) loss	Xq25 region (includes STAG2) loss
Partial monosomy 4p	Partial trisomy 11p	17p12 recurrent (HNPP/CMT1A) region (includes PMP22) gain	Xq28 recurrent region (includes GDI1)
5p15 terminal (Cri du chat syndrome) region gain	Partial monosomy 11q	17p12 recurrent (HNPP/CMT1A) region (includes PMP22) loss	Xq28 recurrent region (int22h1/int22h2-flanked) (includes RAB39B) gain
5q35 recurrent (Sotos syndrome) region (includes NSD1) gain	Partial trisomy 11q	17p13.3 (Miller-Dieker syndrome) region (includes YWHAE and PAFAH1B1) gain	Xq28 recurrent region (int22h1/int22h2-flanked) (includes RAB39B) loss
5q35 recurrent (Sotos syndrome) region (includes NSD1) loss	Partial monosomy 12p	17p13.3 (Miller-Dieker syndrome) region (includes YWHAE and PAFAH1B1) loss	Xq28 region (includes MECP2) gain
			Xq28 region (includes MECP2) loss